

Newton's Laws



Newton's first law and equilibrium

- 1 A book rests motionless on a table. State Newton's first law, and identify the two forces acting on the book that must be balanced.
 - 2 A ball hangs motionless from a single string attached to the ceiling. Identify the two forces acting on the ball and explain why they must be equal and opposite.
 - 3 A crate is pushed at constant velocity across a rough floor. Explain, using Newton's first law, why the applied force must equal the friction force.
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Newton's second law: $F = ma$

- 4 A resultant force of 30 N acts on an object of mass 4 kg. Find its acceleration.
 - 5 A car of mass 1200 kg accelerates from rest to 24 m/s in 8 seconds. Find the resultant force required.
 - 6 An object of mass 6 kg accelerates at 4.5 m/s^2 . Find the resultant force acting on it.
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Weight and mass

- 7 An object has a mass of 12 kg. Find its weight on Earth, using $g = 9.8 \text{ m/s}^2$.
 - 8 An object weighs 88.2 N on Earth. Find its mass, using $g = 9.8 \text{ m/s}^2$.
 - 9 An object has a mass of 25 kg. Find its weight on Earth, using $g = 9.8 \text{ m/s}^2$.
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Multiple forces and the resultant force equation

- 10 A sled of mass 15 kg is pulled across ice by a force of 60 N, while friction of 12 N opposes the motion. Find the sled's acceleration.
- 11 A crate of mass 25 kg is pushed across a floor by a force of 90 N, while friction of 18 N opposes the motion. Find the crate's acceleration.
- 12 A box of mass 10 kg is pushed across a floor by a force of 45 N, while friction of 5 N opposes the motion. Find the box's acceleration.

Newton's third law

- 13** A swimmer pushes backward against the water with a force of 200 N. State Newton's third law, and describe the reaction force in this situation.
- 14** A rocket engine expels exhaust gas downward with a force of 50000 N. State Newton's third law, and describe the reaction force propelling the rocket.
- 15** A person stands on the ground. Identify the action-reaction pair of forces between the person and the Earth, as described by Newton's third law.

Motion in a lift (elevator problems)

- 16** A person of mass 70 kg stands in a lift that accelerates upward at 2 m/s^2 . Find the normal reaction force (apparent weight) from the lift floor, using $g = 9.8 \text{ m/s}^2$.
- 17** The same person now stands in a lift accelerating downward at 1.5 m/s^2 . Find the new normal reaction force.
- 18** A person of mass 60 kg stands in a lift that is moving at constant velocity. Find the normal reaction force from the lift floor, using $g = 9.8 \text{ m/s}^2$.

Connected particles with Newton's second law

- 19** A truck of mass 2000 kg tows a trailer of mass 500 kg using a light rigid tow bar. The engine produces a driving force of 5000 N, and resistance forces of 400 N (truck) and 100 N (trailer) act against the motion. Find the system's acceleration and the tension in the tow bar.
- 20** A car of mass 900 kg tows a trailer of mass 300 kg using a light rigid tow bar. The engine produces a driving force of 2650 N, and resistance forces of 150 N (car) and 100 N (trailer) act against the motion. Find the system's acceleration and the tension in the tow bar.
- 21** A van of mass 1200 kg tows a trailer of mass 300 kg using a light rigid tow bar. The engine produces a driving force of 3300 N, and resistance forces of 200 N (van) and 100 N (trailer) act against the motion. Find the system's acceleration and the tension in the tow bar.

A well-designed word problem: a rocket-powered test sled

- 22 This question has two parts. A test sled of mass 800 kg is launched from rest by a rocket motor that provides a constant thrust of 12000 N. Air and track resistance provide a constant opposing force of 1600 N throughout the motion. (a) Find the sled's acceleration. (b) Find the sled's velocity after it has traveled 50 meters, using $v^2 = u^2 + 2as$.